Price and Quality Comparison Application (PQC)

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Acknowledgment

I would like to express my deepest gratitude to my supervisor, **Dr. Khalid Saleem**, for his exceptional guidance, support, and encouragement throughout the development of this project, ”Price and Quality Comparison Application (PQC).”

His in-depth teaching in Database Management Systems (DBMS) played a vital role in building the technical foundation of this project. The concepts of data normalization, schema design, and efficient querying I learned from him were instrumental in designing the backend architecture and caching mechanisms that power this system. His valuable insights consistently pushed me toward improving the system’s efficiency and reliability.

I would also like to thank all my respected teachers at the Department of Computer Science, Quaid-i-Azam University, who have contributed to my learning journey. Their dedication and knowledge-sharing helped me develop the software engineering skills necessary to complete this work.

This project would not have been possible without the collective mentorship and support I received from my instructors and the university.

Abstract

This project, titled ”Price and Quality Comparison Application (PQC)”, is a web-based information system designed to simplify the shopping experience by aggregating product data from multiple sources. It takes an article’s label and description as input and crawls all possible online and physical store data that are online registered.

The application displays the price and quality ratings of the article along with similar items, allowing users to make informed decisions. A key feature of the system is its ability to correctly process user requests in any language (specifically testing English and Urdu), ensuring a user-friendly interface for a diverse audience. The system leverages web scraping and natural language processing to bridge the gap between online marketplaces and physical store data.

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Chapter 1

Software Project Management Plan (SPMP)

1.1 INTRODUCTION

1.1.1 Project Overview

The Price and Quality Comparison Application (PQC) is a web-based system designed to help users find the best deals on products by comparing prices across various retailers. The system allows users to input a product name (in English or Urdu), translates the query if necessary, and crawls specific target stores (Online and Physical stores that have online store also) to retrieve real-time prices and ratings. Users can view a comparison table that highlights the best price and quality.

1.1.2 Project Deliverables

The deliverables for this project include:

- **Functional Web Application:** A deployed application capable of scraping and displaying prices.
- **User Interface:** A responsive Search and Comparison dashboard built with Next.js.

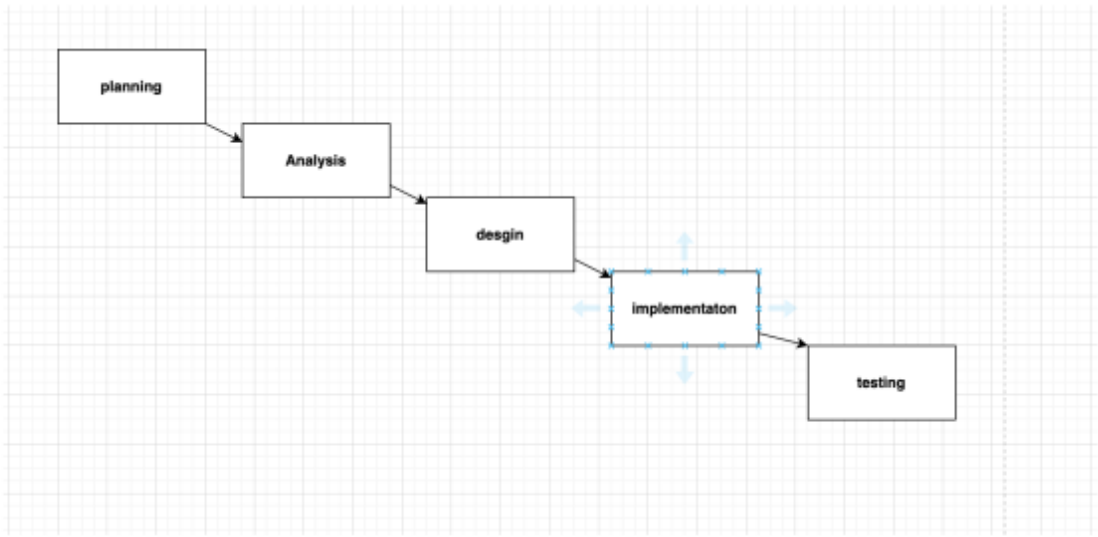
- **Scraping Engine:** A backend system using Playwright to fetch data from Naheed, PriceOye, and Imtiaz.
- **Documentation:** Comprehensive reports including SPMP, SRS, SDD, and STD.

1.2 PROJECT ORGANIZATION

1.2.1 Software Process Model

The Price and Quality Comparison Application (PQC) follows the **Waterfall Model** as its software process model. In this model, development activities are carried out in a sequential manner, where each phase must be completed before the next phase begins. The phases include Planning, Analysis, Design, Implementation, and Testing.

This approach was selected because the project requirements were clearly defined at the beginning, making it suitable to proceed step-by-step with proper documentation and validation at each stage.



1.2.2 Roles and Responsibilities

The roles and responsibilities involved in the development of the PQC system are defined as follows:

- **Developer:** Responsible for application development including coding, database design, user interface implementation, and testing. (Ahmad Masood)
- **Supervisor:** Provides guidance, supervision, and validation of project progress and outcomes. (Dr. Khalid Saleem)
- **User/Admin:** Acts as tester and feedback provider to ensure system functionality and usability. (Ahmad Masood)

1.2.3 Tools and Techniques

The following tools and technologies were used during the development of the project:

- **Programming Languages:** Python, TypeScript and Next.js
- **Database:** PostgreSQL
- **User Interface:** Next.js
- **Web Scraping:** Playwright
- **Translation:** Deep-Translator
- **Development Tools:** Visual Studio Code, Postman, pgAdmin, Cursor, Postgresql

1.3 PROJECT MANAGEMENT PLAN

1.3.1 Tasks

- Planning: MS Word, ProjectLibre
- Analysis: Draw.io, Mermaid.io, MS Word
- Design: MS Word, Draw.io, Mermaid.io
- Database: PostgreSQL

- Development: VS Code, Cursor, Git, GitHub
- Method: Structured Approach

1.3.2 Description

Planning

This includes planning of this software system ”Price and Quality Comparison Application (PQC)” like defining a process model and creating the schedule using ProjectLibre.

Analysis

In analysis we define the requirements, define the use cases, and create use case descriptions.

Design

Design overview, requirement traceability, user interface description, and component description using diagramming tools like Draw.io and Mermaid.io.

Development

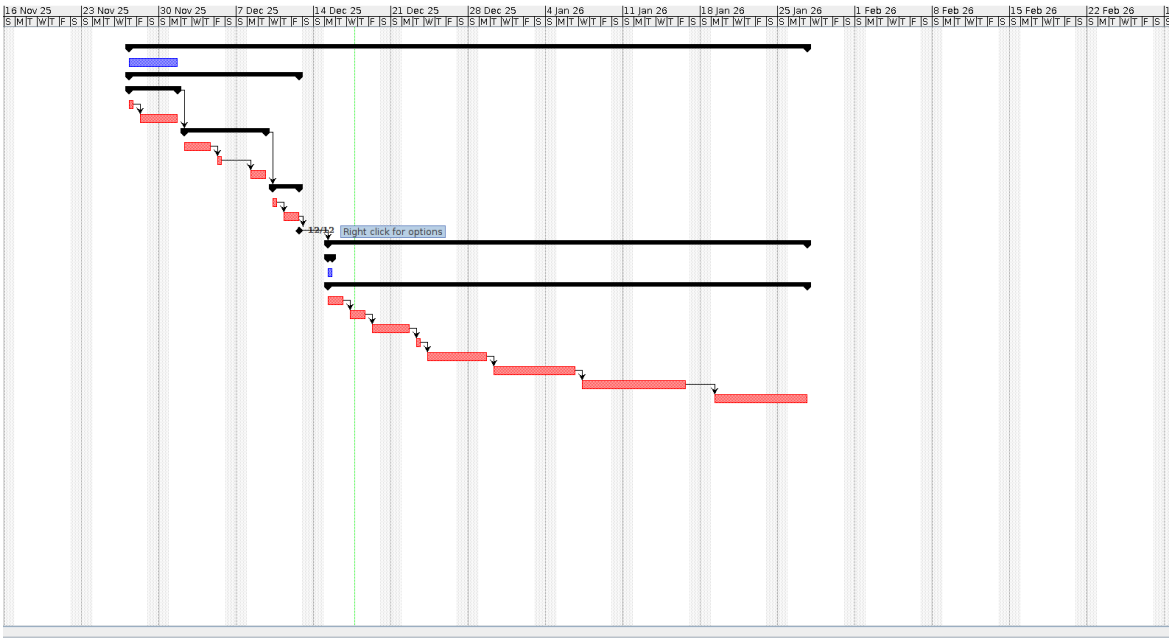
This includes the front (Next.js) and backend (FastAPI) of the system using VS Code and Cursor, with version control managed via Git and GitHub.

Testing

Test cases to check functionality of the system (e.g., verifying price accuracy and Urdu translation).

1.4 Gantt Chart

		Name	Duration	Start	Finish	Predecessors
2		**Project Planning**	**44 days?**	**11/27/25, 8:00 AM**	**1/27/26, 5:00 PM**	
3		Process	3 days?	11/27/25, 8:00 AM	12/1/25, 5:00 PM	
4		**Software Project Management Plan (SPMP)**	**12 days?**	**11/27/25, 8:00 AM**	**12/12/25, 5:00 PM**	
5		**Introduction**	**3 days?**	**11/27/25, 8:00 AM**	**12/1/25, 5:00 PM**	
6		Project overview	1 day?	11/27/25, 8:00 AM	11/27/25, 5:00 PM	
7		Project Deliverables	2 days?	11/28/25, 8:00 AM	12/1/25, 5:00 PM	6
8		**Project Organization**	**6 days?**	**12/2/25, 8:00 AM**	**12/9/25, 5:00 PM**	**5**
9		Software Process Model	3 days?	12/2/25, 8:00 AM	12/4/25, 5:00 PM	
10		Roles and Responsibilities	1 day?	12/5/25, 8:00 AM	12/5/25, 5:00 PM	9
11		Tools and Techniques	2 days?	12/8/25, 8:00 AM	12/9/25, 5:00 PM	10
12		**Project Management Plan**	**3 days?**	**12/10/25, 8:00 AM**	**12/12/25, 5:00 PM**	**8**
13		Tasks	1 day?	12/10/25, 8:00 AM	12/10/25, 5:00 PM	
14		Description	2 days?	12/11/25, 8:00 AM	12/12/25, 5:00 PM	13
15		M1:MileStone Completes	0 days?	12/12/25, 5:00 PM	12/12/25, 5:00 PM	14
16		**Software Requirements Specifications (SRS)**	**32 days?**	**12/15/25, 8:00 AM**	**1/27/26, 5:00 PM**	**15**
17		**Introduction**	**1 day?**	**12/15/25, 8:00 AM**	**12/15/25, 5:00 PM**	
18		Overview	1 day?	12/15/25, 8:00 AM	12/15/25, 5:00 PM	
19		**Requirements**	**32 days?**	**12/15/25, 8:00 AM**	**1/27/26, 5:00 PM**	
20		External Interface Requirements	2 days?	12/15/25, 8:00 AM	12/16/25, 5:00 PM	
21		Functional Requirements	2 days?	12/17/25, 8:00 AM	12/18/25, 5:00 PM	20
22		Non-Functional Requirements	2 days?	12/19/25, 8:00 AM	12/22/25, 5:00 PM	21
23		Database Requirements	1 day?	12/23/25, 8:00 AM	12/23/25, 5:00 PM	22
24		Use Case Diagram	4 days?	12/24/25, 8:00 AM	12/29/25, 5:00 PM	23
25		Use Case description	6 days?	12/30/25, 8:00 AM	1/6/26, 5:00 PM	24
26		System sequence Diagram	8 days?	1/7/26, 8:00 AM	1/16/26, 5:00 PM	25
27		ERD (Entity Relationship Diagram)	7 days?	1/19/26, 8:00 AM	1/27/26, 5:00 PM	26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